

Exam 2025/26 - Basics

- Definition: Random Definitions**
- In this course we only use binary codes, ie  $F = F_2$ , the two-element field
  - For  $a, b \in \mathbb{Z}$ ,  $a$  divides  $b$ , or  $a|b$ , if  $\exists c \in \mathbb{Z}$  s.t.  $b = ac$
  - $a, b \in \mathbb{Z}$  are **relatively prime** or **coprime** if  $\gcd(a, b) = 1$

**Theorem: Random Theorems**

**Well-Ordering Principle:** Every non-empty set of positive integers contains a least element

**Definition: Primitive Pythagorean Triples**

A **Pythagorean Triple** is a set of numbers  $(a, b, c) \in \mathbb{N}$  such that  $a^2 + b^2 = c^2$ . Note if  $(a, b, c)$  is a PT, so is  $(da, db, dc)$ . A **Primitive Pythagorean Triple** is a PT such that  $a, b, c$  have no common factor. Such  $a, b, c$  have one of  $a, b$  odd and the other even. *Square of an odd is odd, square of an even is even. If both even then  $a, b, c$  divisible by 2. If both odd work through the eqn with  $a = 2k + 1, b = 2n + 1$  to see all divisible by 4. Indeed, every PPT with  $(a, b, c)$  with  $a$*

odd has, for  $s \geq t \geq 1$  odd coprime integers,  $a = st, b = \frac{s^2 - t^2}{2}, c = \frac{s^2 + t^2}{2}$ .

**Theorem: The Division Algorithm**

Let  $a, b \in \mathbb{Z}$  with  $b > 0$ . Then  $\exists$  unique  $q, r \in \mathbb{Z}$  s.t.  $a = qb + r$  and  $0 \leq r < b$ .  $q$  is the **quotient** and  $r$  is the **remainder**.

**Lemma: Bézout's Lemma**

Let  $a, b$  be nonzero integers with  $\gcd(a, b)$ . Then  $\exists s, t \in \mathbb{Z}$  s.t.  $as + bt = (a, b)$ . Moreover any common divisor of  $a$  and  $b$  divides  $(a, b)$ .

**Theorem: Fermat's Last Theorem**

For  $n, x, y, z \in \mathbb{Z}$ , and  $n > 2$ , there are no solutions to the equation  $x^n + y^n = z^n$ .

**Theorem: The Extended Euclidean Algorithm**

Let  $a, b \in \mathbb{Z}$  be s.t.  $a \geq b > 0$ . Then  $\gcd(a, b)$  can be found by:

- (1) By the Division Algorithm  $\exists$  unique  $q_1, r_1$  s.t.  $a = q_1 b + r_1$ , with  $0 \leq r_1 < b$
- (2) Assuming  $r_1 > 0$ , we apply the Division Algorithm to  $b, r_1$  for  $b = q_2 r_1 + r_2$ .
- (3) Repeat *ad infinitum*, the last non-zero remainder is  $\gcd(a, b)$

**Example: Finding GCD By Euclidean Algorithm**

Find  $\gcd(112, 27)$ .  
 $112 = 4(27) + 4$     $27 = 6(4) + 3$     $4 = 1(3) + 1$     $3 = 3(1) + 0$   
So  $\gcd(112, 27) = 1$  and 112, 27 are coprime

**Definition: Congruence**

$a$  is **congruent** to  $b \pmod{n}$ , or  $a \cong b \pmod{n}$  means  $n|(a - b)$ , or  $\exists q \in \mathbb{Z}$  s.t.  $a = nq + b$ .  
For  $a, b, c, d, n \in \mathbb{Z}$  with  $n > 0$ ,  $a \cong b \pmod{n}$ ,  $c \cong d \pmod{n}$ :

- $a + c \cong b + d \pmod{n}$
- $a - c \cong b - d \pmod{n}$
- $ac \cong bd \pmod{n}$

**Definition: System Of Residues**

A **complete system of residues modulo**  $n$  is a set  $I \subseteq \mathbb{Z}$  s.t. every  $k \in \mathbb{Z}$  is congruent  $\pmod{n}$  to exactly one  $i \in I$

**Definition: The Ring  $\mathbb{Z}/n\mathbb{Z}$  or  $\mathbb{Z}_n$**

The elements of  $\mathbb{Z}_n$  are congruence classes  $\pmod{n}$ , ie  $\mathbb{Z}_n = \{\bar{0}, \bar{1}, \dots, \overline{n-1}\}$ . We define the ring addition/multiplication by, for  $0 \leq a, b < n$ :

- $\bar{a} + \bar{b} = \bar{c}$  iff  $a + b = c \pmod{n}$
- $\bar{a} - \bar{b} = \bar{c}$  iff  $a - b = c \pmod{n}$
- $\bar{a} \cdot \bar{b} = \bar{c}$  iff  $a \cdot b = c \pmod{n}$

**Theorem: Wilson's Theorem**

If  $p$  is prime then  $(p-1)! \cong -1 \pmod{p}$

**Major Theorem: The Chinese Remainder Theorem**

Let  $m_1, m_2, \dots, m_k \in \mathbb{N}$  be pairwise coprime. Then given  $a_1, a_2, \dots, a_k \in \mathbb{Z}$ , the system of congruences

$$\begin{aligned} x &\cong a_1 \pmod{m_1} \\ x &\cong a_2 \pmod{m_2} \\ &\vdots \\ x &\cong a_k \pmod{m_k} \end{aligned}$$

has a solution which is unique  $\pmod{N}$ , where  $N = m_1 \cdot m_2 \cdot \dots \cdot m_k$ . The proof is constructive:

*Proof.* Let  $N_i = n/m_i$  be the product of all  $m_j$  with  $j \neq i$ . Since  $m_i$  is coprime to  $m_j$  for  $i \neq j$ , we must have  $\gcd(N_i, m_i) = 1$ . Then since  $N_i$  and  $m_i$  are coprime,  $N_i$  must have an inverse  $\pmod{m_i}$ , ie  $\exists x_i \in \mathbb{Z}$  s.t.  $N_i x_i \cong 1 \pmod{m_i}$  (to find  $x_i$  use the Euclidean algorithm to find numbers s.t.  $N_i x_i m_i y_i = 1$ ). Then consider the product  $a_i N_i x_i$ , which is s.t.  $a_i N_i x_i \cong 0 \pmod{m_j}$  for  $i \neq j$ , and  $a_i N_i x_i \cong a_i \pmod{m_i}$ . Thus the sum  $x = \sum_{i=1}^k a_i N_i x_i$  precisely satisfies  $x \cong a_i \pmod{m_i}$  for all  $i$ .  $\square$

**Example: Applying The CRT**

Solve the system of congruences

$$\begin{aligned} x &\cong 1 \pmod{2} \\ x &\cong 3 \pmod{5} \\ x &\cong 4 \pmod{7} \end{aligned}$$

We have  $N = 2 \cdot 5 \cdot 7 = 70$ ,  $N_1 = 5 \cdot 7 = 35$ ,  $N_2 = 2 \cdot 7 = 14$ ,  $N_3 = 2 \cdot 5 = 10$ . Calculate by Euclidean Algorithm or inspection for:

$$\begin{aligned} 35 \cdot 1 &\cong 1 \pmod{2} \\ 14 \cdot 4 &\cong 1 \pmod{5} \\ 10 \cdot 5 &\cong 1 \pmod{7} \end{aligned}$$

so we have  $x_1 = 1, x_2 = 4, x_3 = 5$ .  
Thus  $x \cong (1 \cdot 35 \cdot 1) + (3 \cdot 14 \cdot 4) + (4 \cdot 10 \cdot 5) \cong 403 \cong 53 \pmod{70}$ .

**Theorem: Applying The CRT For Non-Coprime  $m_i m_j$**

Key fact - if  $m, n$  are coprime integers, and  $a \in \mathbb{Z}$ , the congruence  $x \cong a \pmod{m \cdot n}$  has the same integer solutions as the system

$$\begin{aligned} x &\cong a \pmod{m} \\ x &\cong a \pmod{n} \end{aligned}$$

Thus knowing  $x \pmod{N}$  is equivalent to knowing  $x$  modulo each of the primes in the prime decomposition of  $N$ , and to solve such systems we can simply decompose it to a larger system where all  $m_i$  are coprime.

Binary Codes

**Major Definition: Binary Codes**

A **binary word** is a sequence of 1s and 0s, with the **word length** the total number of digits in the word. A **binary code** is a set of binary words.

- Binary words which are elements of a binary code are **codewords**. We can ascribe meaning to codewords (eg  $(1, 1, 1) =$  'yes' and  $(0, 0, 1) =$  'no').
- A code is a **block code** if all words have the same length.
- A code  $C$  is **linear** iff the sum of any  $y, z \in C$  is also in  $C$ .
- A **repetition code of length  $n$**  is obtained by repeating  $n - 1$  times the first single symbol. Example of length 3:  $\{(1, 1, 1), (0, 0, 0)\}$ .
- A binary linear code has exactly  $2^k$  elements.
- Binary linear codes can be considered as linear spaces over  $\mathbb{F} = \{0, 1\}$ .
- $x \cdot y := \sum_{i=1}^n x_i y_i$ .  $(\cdot)$  is a symmetric bilinear form on  $\mathbb{F}_2^n$ .
- $x$  and  $y$  are **orthogonal** if  $x \cdot y = 0$ .
- A **parity check bit** is a bit appended to a codeword to make the sum of the digits even. All codewords with parity check bits are of form  $(x_1, \dots, x_{n-1}, \sum_{i=1}^{n-1} x_i)$

**Definition: Neighbourhood Of A Word**

For a binary word  $x = (x_1, \dots, x_n)$ , with all  $x_i \in \{0, 1\}$  and  $r \geq 0$ , the **neighbourhood of  $x$  with radius  $r$**  is the set  $N_r(x) = \{y = (y_1, \dots, y_n) : y_1, \dots, y_n \in \{0, 1\}, d(x, y) \leq r\}$ .

- Note that  $x \in N_r(x)$
- $|N_r(x)| = \binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{r}$

**Definition: The Parameters Of A Linear Code**

A triple  $(n, k, d)$  gives the three params of a linear code -  $n$ , the length of codewords in the code;  $k$ , the dimension of the code  $\dim_{\mathbb{F}} C$ , and  $d$  the minimal distance of the code.

- $k$  is equal to the number of linearly independent codewords in the code

**Definition: Hamming Distance**

For  $C$  a code over  $F_2$ ,  $x = (x_1, x_2, \dots, x_n), y = (y_1, y_2, \dots, y_n) \in C$ ; the **Hamming distance  $d$  on  $C$**  is  $d(x, y) := |\{i \in \{1, \dots, n\} : x_i \neq y_i\}|$

- For  $x \in C$ , the **weight** of  $x$  is  $\text{wt}(x) := d(x, 0) = |\{i \in \{1, \dots, n\} : x_i \neq 0\}|$
- The **minimal distance** of  $C$  is  $d(C) := \min\{d(x, y) : x, y \in C, x \neq y\}$ , intuitively the smallest distance between any two codewords in the code.
- For  $C$  a linear code, the minimal distance is equal to the minimum weight among non-zero codewords in the code.
- For binary words  $x, y, z$  of length  $n$ ;  $d(x, y) + d(y, z) \geq d(x, z)$

**Definition: Decision Rules**

For  $C \subseteq F^n$  a code, a **decision rule** for  $C$  is a function  $\sigma : F^n \rightarrow C$  which assigns to each  $z \in F^n$  a codeword  $c \in C$ .

**Definition: Bit-Error**

A **bit-error** is a mistake in transmission of a codeword  $z$  in a code  $C$ , where one digit of  $z$  is flipped (either from 0 to 1, or from 1 to 0).

**Theorem: Minimal Distance Rule**

For  $C$  a code, the **minimal distance rule** states that given a codeword  $z \in C$  received through a noisy channel, the receiver should choose  $\sigma(z)$  to be the codeword  $c \in C$  such that  $d(z, c)$  is minimised. If a sender uses a code  $C$  with minimum distance  $d > 2r$  and the receiver uses the MD rule, provided no more than  $r$  bit-errors are made in transmitting any codeword, the MD rule will map every received codeword to the correct transmitted one.

**Definition: A Generator Matrix Of A Code**

Let  $C$  be a linear code with params  $(n, k, d)$ . Since  $C$  has  $\dim_{\mathbb{F}} C = k$  we can find codewords  $c_1, \dots, c_k$  which form a basis of  $C$ . Then the  $k \times n$  matrix with rows  $c_1, \dots, c_k$  is a **generator matrix** of  $C$

**Definition: The Dual Code**

For  $C$  a code with params  $(n, k, d)$ , the **dual code**  $C^\perp = \{x = (x_1, \dots, x_n) : x_1, \dots, x_n \in \mathbb{F}_2, x \cdot c = 0 \forall c \in C\}$  is the orthogonal complement of  $C$ .

- $\dim_{\mathbb{F}} C^\perp = n - k$  since  $\dim_{\mathbb{F}} C = k$ .
- A code is **self-dual** if  $C = C^\perp$ , for example  $C = \{(0, 0), (1, 1)\}$ .
- $(C^\perp)^\perp = C$ .
- $\dim C + \dim C^\perp = n$ .

**Definition: Parity Check Matrices**

For  $C$  a code with params  $(n, k, d)$ , a generator matrix  $H$  for  $C^\perp$  (a matrix whose rows form a basis of  $C^\perp$ ) is a **parity check matrix** for  $C$ .

- $H$  is a  $(n - k) \times n$  matrix
- $c$  is a codeword in  $C$  iff  $Hc^T = 0$  iff  $cH^T = 0$ .
- $C$  has minimal distance  $d$  iff  $H$  has  $d$  lin-dep cols and does not have  $d - 1$  lin-dep cols

**Definition: Perfect Codes**

Let  $C$  be a binary code of length  $n$  and minimal distance  $d = 2r + 1$  for integer  $r \geq 0$ . Then  $C$  is a **perfect code** iff  $|C|(\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{r}) = 2^n$

- Equivalent def - if each binary codeword  $c \in C$  of length  $n$  is in exactly one neighbourhood  $N_r(c)$
- Equivalent def - if  $d \geq 1$ ,  $C$  is a perfect code iff  $\exists r$  s.t. each binary word of length  $n$  is in exactly one neighbourhood  $N_r(c)$  where  $c$  is a codeword in  $C$ .

Euler & Rings

**Major Definition: Euler's Function**

For  $n > 1$  Euler function  $\phi(n)$  is defined to be the number of natural numbers not exceeding  $n$  which are coprime with  $n$ , and  $\phi(1) := 1$ .

**Major Theorem: Euler's Theorem**

For  $n \in \mathbb{N}$  s.t.  $n > 1$ , and  $a$  any number s.t.  $a$  and  $n$  are coprime. Then  $a^{\phi(n)} - 1 \cong 0 \pmod{n}$